Experiment 4

Date : 02-02-2026

Name : M.Sreekar
Roll no : 31
Reg : 127156098

Linear Regression

Description :

Linear regression algorithm shows a linear relationship between a dependent (y) and one or more independent (y) variables, hence called linear regression. The linear regression model provides a sloped straight line representing the relationship between the variables.

Algorithm :

Step 1: Decide on purpose of model and appropriate dependent variable to meet that purpose.

Step 2: Decide on independent variables.

Step 3: Estimate parameters of regression equation.

Step 4: Interpret estimated parameters, goodness of fit and qualitative and quantitative assessment of parameters.

Step 5: Assess appropriateness of assumptions.

Step 6: If some assumptions are not satisfied, modify and revise estimated equation.

Step 7: Validate estimated regression equation.

Dataset : https://archive.ics.uci.edu/ml/machine-learning-databases/balan ce-scale/balance-scale.data

Dataset Information :

This data set was generated to model psychological experimental results. Each example is classified as having the balance scale tip to the right, tip to the left, or be balanced. The attributes are the left weight, the left distance, the right weight, and the right distance. The correct way to find the class is the greater of (left-distance * left-weight) and (right-distance * right-weight). If they are equal, it is balanced.

No.of instances: 625

No.of attributes: 4

Dataset Information :

1. Class Name: 3 (L, B, R)
2. Left-Weight: 5 (1, 2, 3, 4, 5)
3. Left-Distance: 5 (1, 2, 3, 4, 5)
4. Right-Weight: 5 (1, 2, 3, 4, 5)
5. Right-Distance: 5 (1, 2, 3, 4, 5)

Ex - 4- Linear Regression

Code - LR: & MLR

LINEAR REGRESSION

Description:

Linear regression algorithm shows a linear relationship between a dependent (y) and one or more independent (y) variables, hence called linear regression. The linear regression model provides a sloped straight line representing the relationship between the variables.

Algorithm:

Step 1: Decide on purpose of model and appropriate dependent variable to meet that purpose.

Step 2: Decide on independent variables.

Step 3: Estimate parameters of regression equation.

Step 4: Interpret estimated parameters, goodness of fit and qualitative and quantitative assessment of parameters.

Step 5: Assess appropriateness of assumptions.

Step 6: If some assumptions are not satisfied, modify and revise estimated equation.

Step 7: Validate estimated regression equation.

Dataset:

https://archive.ics.uci.edu/ml/machine-learning-databases/cpu-performance/machine.data

Step : 0 - Import the necessary libraries

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt
```

Step : 1 - Load the dataset

```
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/cpu-performance/machine.data"

col_names = ['vendor_name', 'model_name', 'MYCT', 'MMIN', 'MMAX', 'CACH',
```

```
             'CHMIN', 'CHMAX', 'PRP', 'ERP']

data = pd.read_csv(url,header=None,names=col_names)

(data).head()
```


```
{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"vendor_name","rawType":"object","type":"string"},
{"name":"model_name","rawType":"object","type":"string"},
{"name":"MYCT","rawType":"int64","type":"integer"},
{"name":"MMIN","rawType":"int64","type":"integer"},
{"name":"MMAX","rawType":"int64","type":"integer"},
{"name":"CACH","rawType":"int64","type":"integer"},
{"name":"CHMIN","rawType":"int64","type":"integer"},
{"name":"CHMAX","rawType":"int64","type":"integer"},
{"name":"PRP","rawType":"int64","type":"integer"},
{"name":"ERP","rawType":"int64","type":"integer"}],"ref":"30e478cd-
4d00-4083-9f3b-01ecb2fd019e","rows":
[["0","adviser","32/60","125","256","6000","256","16","128","198","199
"],["1","amdahl","470v/
7","29","8000","32000","32","8","32","269","253"],
["2","amdahl","470v/
7a","29","8000","32000","32","8","32","220","253"],
["3","amdahl","470v/
7b","29","8000","32000","32","8","32","172","253"],
["4","amdahl","470v/
7c","29","8000","16000","32","8","16","132","132"]],"shape":
{"columns":10,"rows":5}}
```


```
print("Dataset Information:")
print(data.info())
print("\nMissing values:")
print(data.isnull().sum())
print("\nBasic statistics:")
print(data.describe())
print("\nUnique vendors:", data['vendor_name'].nunique())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 209 entries, 0 to 208
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   vendor_name  209 non-null    object
 1   model_name   209 non-null    object
 2   MYCT         209 non-null    int64
 3   MMIN         209 non-null    int64
 4   MMAX         209 non-null    int64
 5   CACH         209 non-null    int64
 6   CHMIN        209 non-null    int64
 7   CHMAX        209 non-null    int64
```

```
Unique vendors: 30
```

Step 1: Purpose of the Model Purpose: To predict the "peak performance" (or another relevant feature) of a CPU based on other features.

Dependent Variable: Choose "peak performance" (feature 10, based on the dataset description).

Purpose: To predict the "peak_performance" of a CPU based on its hardware specifications.

Dependent Variable (y): "peak_performance" - This is the target variable we want to predict.

Independent Variables (X): We'll use numerical features that likely influence CPU performance:

- motherboard_manufacturer
- cpu_manufacturer

- clock_speed
- bus_speed
- cache_size
- branch_prediction
- branch_mispredictions
- instructions_per_cycle

We'll exclude vendor_name and model_name as they are categorical identifiers.

```
features = ['MYCT', 'MMIN', 'MMAX', 'CACH', 'CHMIN', 'CHMAX']
X = data[features]
y = data['PRP']

print(f"Features shape: {X.shape}")
print(f"Target shape: {y.shape}")
print(f"\nTotal samples: {len(data)}")
print(f"\nTarget (PRP) statistics:")
print(f"Mean: {y.mean():.2f}")
print(f"Min: {y.min()}")
print(f"Max: {y.max()}")
print(f"Std: {y.std():.2f}")
```

```
Features shape: (209, 6)
Target shape: (209,)

Total samples: 209

Target (PRP) statistics:
Mean: 105.62
```

```
Min: 6
Max: 1150
Std: 160.83
```

```
# Step 7: Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42, shuffle=True)

print(f"Training set shape: {X_train.shape}")
print(f"Testing set shape: {X_test.shape}")
print(f"\nTraining target statistics:")
print(f"Mean: {y_train.mean():.2f}, Std: {y_train.std():.2f}")
print(f"\nTesting target statistics:")
print(f"Mean: {y_test.mean():.2f}, Std: {y_test.std():.2f}")
```

```
Training set shape: (167, 6)
Testing set shape: (42, 6)

Training target statistics:
Mean: 99.45, Std: 139.07

Testing target statistics:
Mean: 130.17, Std: 228.36
```

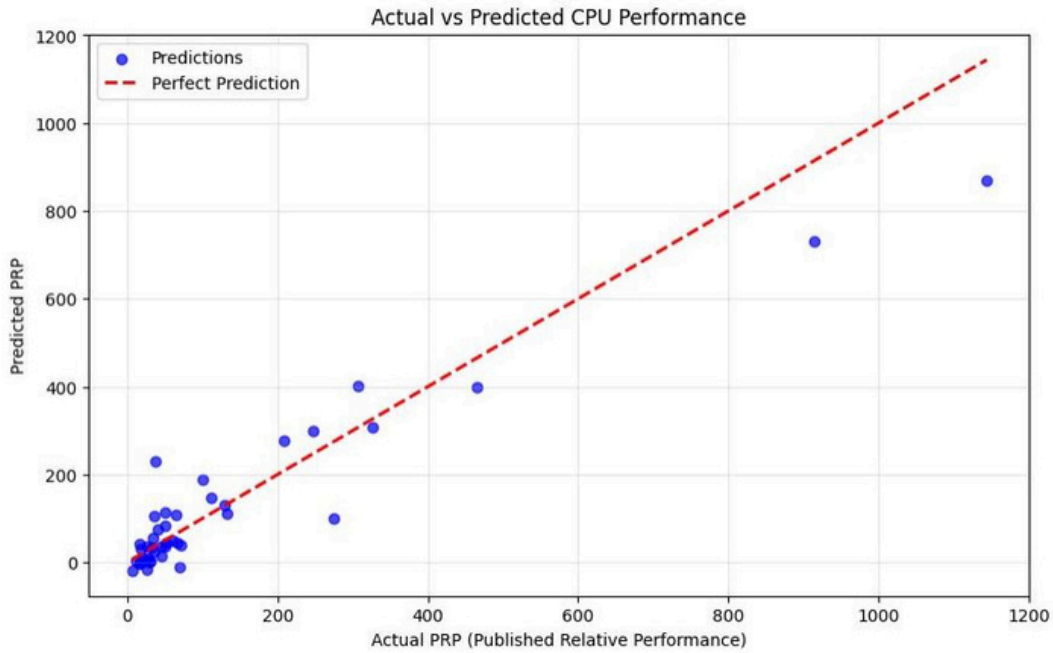
```
# Step 8: Build Linear Regression Model
# Add constant for intercept term
X_train_const = sm.add_constant(X_train)
X_test_const = sm.add_constant(X_test)

# Create and fit the model
model = sm.OLS(y_train, X_train_const)
results = model.fit()

print("="*60)
print("LINEAR REGRESSION MODEL RESULTS")
print("="*60)
print(results.summary())
```

```
============================================================
LINEAR REGRESSION MODEL RESULTS
============================================================
                            OLS Regression Results

==============================================================================
========
Dep. Variable:                    PRP   R-squared:
0.838
Model:                            OLS   Adj. R-squared:
0.832
Method:                 Least Squares   F-statistic:
```

```
there are
strong multicollinearity or other numerical problems.

# Step 9: Make predictions on test set
y_pred = results.predict(X_test_const)

# Calculate evaluation metrics
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("="*60)
print("MODEL EVALUATION METRICS")
print("="*60)
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.2f}")
print(f"R-squared (R²) Score: {r2:.4f}")
print(f"\nModel explains {r2*100:.2f}% of the variance in CPU
performance")
```

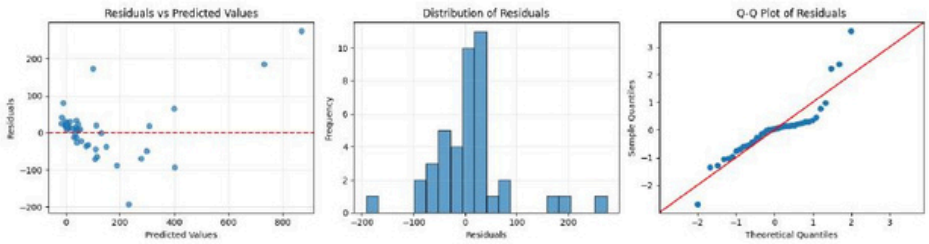
```
============================================================
MODEL EVALUATION METRICS
============================================================
Mean Squared Error (MSE): 5633.07
Root Mean Squared Error (RMSE): 75.05
R-squared (R²) Score: 0.8893

Model explains 88.93% of the variance in CPU performance
```

```
# Step 10: Visualize Actual vs Predicted values
plt.figure(figsize=(10, 6))
plt.scatter(y_test, y_pred, alpha=0.7, color='blue',
label='Predictions')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
         'r--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual PRP (Published Relative Performance)')
plt.ylabel('Predicted PRP')
plt.title('Actual vs Predicted CPU Performance')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```




```
# Step 11: Analyze residuals
residuals = y_test - y_pred

plt.figure(figsize=(15, 4))

# Plot 1: Residuals vs Predicted
plt.subplot(1, 3, 1)
plt.scatter(y_pred, residuals, alpha=0.7)
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel('Predicted Values')
plt.ylabel('Residuals')
plt.title('Residuals vs Predicted Values')
plt.grid(True, alpha=0.3)

# Plot 2: Histogram of residuals
plt.subplot(1, 3, 2)
plt.hist(residuals, bins=20, edgecolor='black', alpha=0.7)
plt.xlabel('Residuals')
plt.ylabel('Frequency')
plt.title('Distribution of Residuals')
plt.grid(True, alpha=0.3)

# Plot 3: Q-Q plot for normality
plt.subplot(1, 3, 3)
sm.qqplot(residuals, line='45', fit=True, ax=plt.gca())
plt.title('Q-Q Plot of Residuals')

plt.tight_layout()
```

```
plt.show()

print(f"Residual Statistics:")
print(f"Mean: {residuals.mean():.4f}")
print(f"Std: {residuals.std():.4f}")
print(f"Min: {residuals.min():.4f}")
print(f"Max: {residuals.max():.4f}")
```



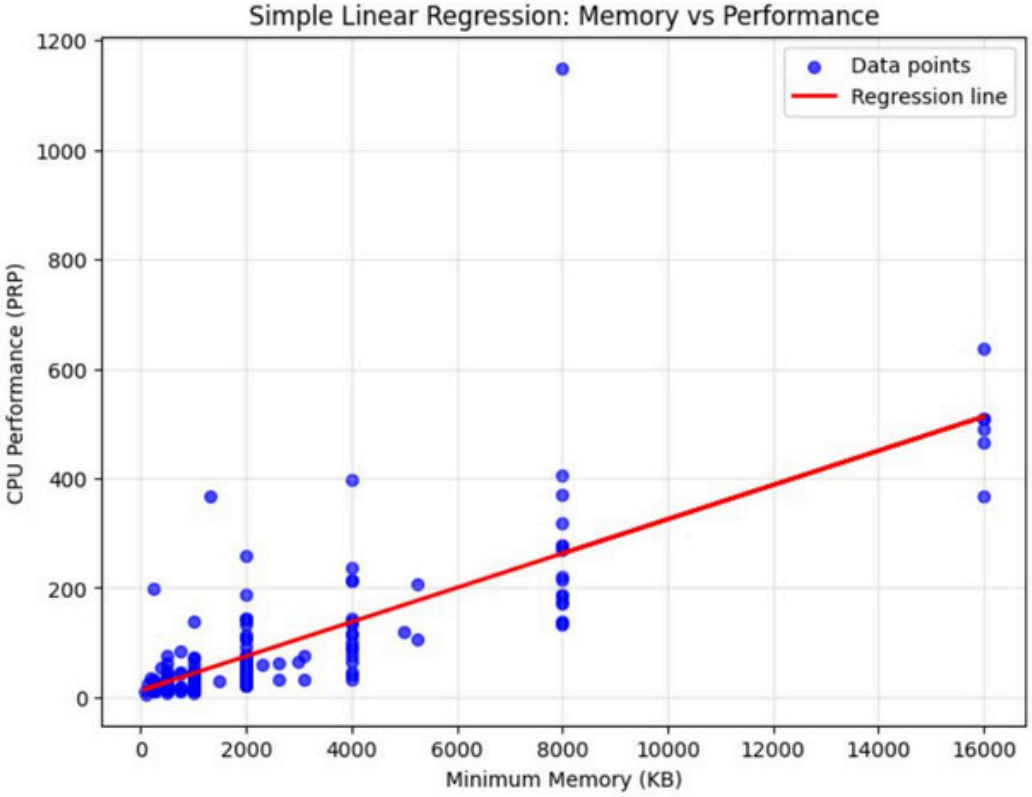

```
Residual Statistics:
Mean: 7.5719
Std: 75.5760
Min: -192.3930
Max: 274.4820

# Step 12: Interpret significant features
print("="*60)
print("SIGNIFICANT FEATURES ANALYSIS (p < 0.05)")
print("="*60)

significant_features = results.pvalues[results.pvalues < 0.05]
print(f"Number of significant features: {len(significant_features) -
1}")

print("\nSignificant features with coefficients:")
for feature, pval in significant_features.items():
    if feature != 'const':
        coef = results.params[feature]
        print(f"{feature:10s}: Coefficient = {coef:8.4f}, p-value =
{pval:.6f}")

print(f"\nModel Intercept: {results.params['const']:.4f}")

============================================================
SIGNIFICANT FEATURES ANALYSIS (p < 0.05)
============================================================
Number of significant features: 5

Significant features with coefficients:
MYCT      : Coefficient =   0.0428, p-value = 0.020501
MMIN      : Coefficient =   0.0111, p-value = 0.000002
MMAX      : Coefficient =   0.0060, p-value = 0.000000
```

```
CACH      : Coefficient =   0.6213, p-value = 0.000031
CHMAX     : Coefficient =   1.3829, p-value = 0.000000

Model Intercept: -49.6937
```

```
# Step 13: Simple Linear Regression Example (Manual Calculation)
def estimate_coef(x,y) :
    #number of obs/pts
    n = np.size(x)

    #mean of x and y vector
    m_x = np.mean(x)
    m_y = np.mean(y)

    #calculating cross-deviation and devviation about x

    SS_xy = np.sum(y*x) - n*m_y*m_x
    SS_xx = np.sum(x*x) - n*m_x*m_x

    # calculating regression const
    b_1 = SS_xy / SS_xx
    b_0 = m_y - b_1*m_x

    return(b_0,b_1)

def plot_regression_line(x, y, b, x_label='x', y_label='y',
title='Linear Regression'):
    """Plot regression line"""
    plt.figure(figsize=(8, 6))
    plt.scatter(x, y, color="blue", marker="o", s=30, alpha=0.7,
label='Data points')

    # Predicted values
    y_pred = b[0] + b[1] * x

    # Plot regression line
    plt.plot(x, y_pred, color="red", linewidth=2, label='Regression
line')

    plt.xlabel(x_label)
    plt.ylabel(y_label)
    plt.title(title)
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.show()

    return y_pred

# Step 14: Example of simple linear regression with one feature
print("="*60)
```

```python
print("SIMPLE LINEAR REGRESSION: MMIN vs PRP")
print("="*60)

# Use minimum memory (MMIN) as predictor
x_simple = X_train['MMIN'].values
y_simple = y_train.values

# Estimate coefficients manually
b_simple = estimate_coef(x_simple, y_simple)
print(f"Manual calculation results:")
print(f"Intercept (b_0): {b_simple[0]:.4f}")
print(f"Slope (b_1): {b_simple[1]:.4f}")
print(f"\nInterpretation: For every 1KB increase in minimum memory,")
print(f"CPU performance increases by {b_simple[1]:.4f} units")

# Plot
y_pred_simple = plot_regression_line(
    x_simple, y_simple, b_simple,
    x_label='Minimum Memory (KB)',
    y_label='CPU Performance (PRP)',
    title='Simple Linear Regression: Memory vs Performance'
)

# Calculate R²
ss_res = np.sum((y_simple - y_pred_simple) ** 2)
ss_tot = np.sum((y_simple - np.mean(y_simple)) ** 2)
r2_simple = 1 - (ss_res / ss_tot)
print(f"\nR² for simple regression: {r2_simple:.4f}")
print(f"Simple model explains {r2_simple*100:.2f}% of variance")
```

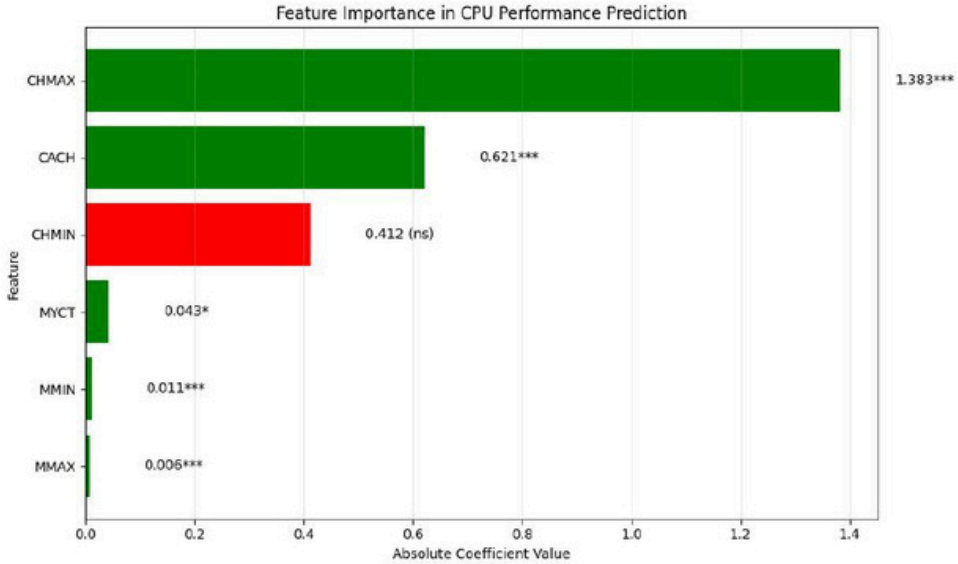
```
============================================================
SIMPLE LINEAR REGRESSION: MMIN vs PRP
============================================================
Manual calculation results:
Intercept (b_0): 12.3783
Slope (b_1): 0.0313

Interpretation: For every 1KB increase in minimum memory,
CPU performance increases by 0.0313 units
```




```
R² for simple regression: 0.5761
Simple model explains 57.61% of variance
```

```
# Step 15: Feature importance visualization
feature_importance = pd.DataFrame({
    'Feature': features,
    'Coefficient': [results.params[f] for f in features],
    'Absolute_Coef': [abs(results.params[f]) for f in features],
    'P_value': [results.pvalues[f] for f in features]
})

# Sort by absolute coefficient value
feature_importance = feature_importance.sort_values('Absolute_Coef',
ascending=False)

plt.figure(figsize=(10, 6))
colors = ['green' if p < 0.05 else 'red' for p in
feature_importance['P_value']]
plt.barh(feature_importance['Feature'],
feature_importance['Absolute_Coef'], color=colors)
plt.xlabel('Absolute Coefficient Value')
plt.ylabel('Feature')
plt.title('Feature Importance in CPU Performance Prediction')
```

```
plt.grid(True, alpha=0.3, axis='x')
plt.axvline(x=0, color='black', linewidth=0.8)

# Add significance labels
for i, (coef, pval) in
enumerate(zip(feature_importance['Coefficient'],
feature_importance['P_value'])):
    significance = '***' if pval < 0.001 else '**' if pval < 0.01 else
'*' if pval < 0.05 else ' (ns)'
    plt.text(abs(coef) + 0.1, i, f'{coef:.3f}{significance}',
va='center')

plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()

print("\nFeature Importance Summary:")
print(feature_importance.to_string(index=False))
```




```
Feature Importance Summary:
Feature  Coefficient  Absolute_Coef      P_value
  CHMAX     1.382874       1.382874 1.151493e-07
   CACH     0.621287       0.621287 3.104730e-05
  CHMIN     0.411682       0.411682 7.037944e-01
   MYCT     0.042756       0.042756 2.050100e-02
   MMIN     0.011096       0.011096 2.239731e-06
   MMAX     0.005968       0.005968 3.427878e-15
```